

3 INNOVATIVE DRONE TECHNOLOGIES

FROM IG DRONES DRIVING EFFICIENCY ACROSS SECTORS

Haryana-based IG Drones develops advanced surveillance, agriculture, and defence technologies, focusing on improving adaptability and operational precision.

Founded in 2018 by Om Prakash and Bodhisattwa Sanghapriya, IG Drones is a drone development company specialising in defence, industrial, agricultural, and skilling applications.

In the industrial sector, the IG Drones Jaga facilitates surveillance and mapping with high precision. For agriculture, the IG Drones Kisan provides efficient alternatives to manual farming practices by spraying pesticides and fertilisers. In defence, the Delta 400, a hand-launched fixed-wing drone requiring a runway for landing, is suited for extended flight missions. The Skyhawk VTOL, capable of vertical take-off and landing, operates effectively in challenging terrains, making it suitable for critical defence applications. These drones address specific sectoral needs, enhancing both efficiency and adaptability.

“Beyond drones, we offer anti-drone systems to counter emerging threats. These include handheld anti-drone guns and fixed installations equipped for 3D perimeter scans, ensuring area security. Our systems also address non-drone threats, such as road-based risks, providing comprehensive security solutions,” explains Sambit Prasad Parida, Co-founder and CTO of IG Drones.

Sambit further elaborates on the company name IG Drones, which is derived from its parent organisation, InventGrid India Private Limited. The ‘I’ represents ‘Invent’ while

the ‘G’ signifies ‘Grid’ reflecting the company’s foundational principles.

Headquartered in Noida, IG Drones operates through five regional offices strategically located across India. The company’s leadership team includes founders Bodhisattwa and Om Prakash and co-founders Sambit Prasad Parida, Shuvam Dash, and Ashish Kumar Jena. With over 150 employees, IG Drones manages diverse operations focusing on innovation and growth.

The company manufactures airframes, motors, and avionics components from its facilities in Rajkot and Baroda. It operates an in-house lab for composite manufacturing, using chambering machines and layout process tools. Motor production involves winding copper around poles with precision winding machines. While electronics manufacturing is outsourced, the company designs components, sources parts externally, and completes assembly using pick-and-place machines. Each of its five production lines can produce five units per week.

IG Drones holds design patents and has pioneered various applications and subcomponent technologies. These include advanced holding mechanisms, optimised component placement, and patented propeller designs that minimise noise at specific altitudes. Its drones also feature sensors like thermal and near-infrared cameras for wind blade inspections, enabling the detection of hidden damage.



Sambit P., CTO and Co-Founder, IG Drones

Additionally, specialised components facilitate contact testing for detailed analysis. IG Drones has also developed India’s first 5G-enabled drone and an indigenously designed simulator.

The company’s revenue model includes drone sales, Drones as a Service, and Software as a Service. Under the Drones as a Service model, customers purchase drones and access supplementary services to maximise operational value.

IG Drones is increasing its research and development efforts to support the army, focusing on medium- and high-altitude drones to reduce reliance on foreign partners. The company is also exploring unmanned surface vehicles with underwater capabilities to complement its aerial systems, addressing a broader spectrum of operational requirements. **EFY**

—Nitisha Dubey